1 Introduction (Option 3: Teleportation)

The top 10 Marvel characters with the highest PageRank values ($\beta = 0.85$):

1. SPIDER-MAN: 0.00548
2. CAPT. AMERICA: 0.00527
3. IRON MAN: 0.00439
4. WOLVERINE: 0.00403
5. THING: 0.00402
6. MR. FANTASTIC: 0.00391
7. HUMAN TORCH: 0.00385
8. SCARLET WITCH: 0.00366
9. THOR: 0.00360
10. BEAST: 0.00355

1.1 Hypothesis

Lower β values lead to more random jumps, diffusing importance across the graph and typically resulting in lower individual PageRank scores for top nodes. Higher β instead prioritize actual links in the graph allowing actual central nodes to accumulate more PageRank value instead of losing it from random jumps

1.2 Metrics

We use a Position-Aware Weighted Jaccard similarity to compare stability across β shifts:

$$J_{pos}(L_A, L_B, \lambda) = \frac{\sum_{x \in S_A \cap S_B} \exp(-\lambda \cdot |pos_A(x) - pos_B(x)|)}{|S_A \cup S_B|}$$

This measure evaluates the similarity between PageRank results from different β values by accounting for both the presence of items and their rank stability. The further an item position moves the less it contributes to the overall similarity

2 Experiment

We executed PageRank on the Marvel dataset across β values ranging from $0 \to 1$ in increments of 0.05.

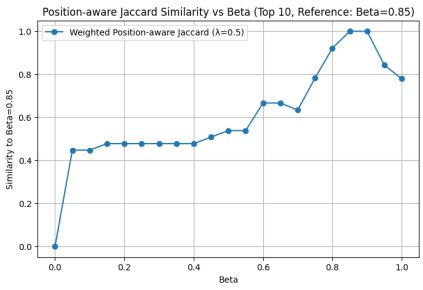
3 Conclusion

As observed in plot 1a the weighted Jaccard similarity computed between each β value and a reference of $\beta = 0.85$ for the top-10 ranks, we can see that it's value remains relatively stable for lower β values. This is also confirmed by 1b (comparing neighboring β's). In 1c we can also see that for lower β's their PageRank values are lower.

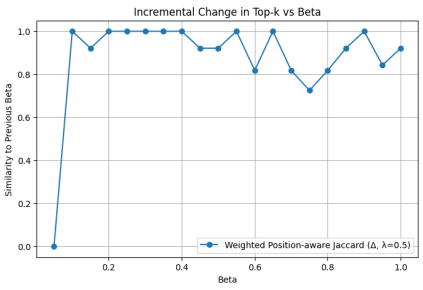
This makes sense since, by allowing more random jumps, the PageRank scores are more evenly distributed. This results in more compressed and less differentiated PageRank values meaning that there is more inherent uncertainty in the results.

For larger β values, instead, we observe more rank changes, in fact adhering more to the input graph means allowing the more important nodes to accumulate more PageRank value, resulting in more spread out PageRank scores for the top-10 nodes leading to a more strongly defined hierarchy.

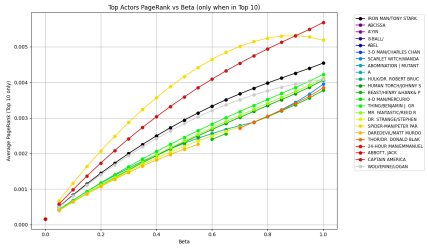
We can also observe that some PageRank are decreasing in value while approaching 1, this is due to the fact that the algorithm adheres almost completely with the input graph, concentrating on pockets of the graph while draining others. An example is "CAPTAIN AMERICA" overtaking "SPIDER-MAN" at high β values. We can notice that "CAPTAIN AMERICA" contains a self-loop that absorb PageRank without teleportation (high β's), starving other nodes like "SPIDER-MAN"



(a) Weighted Jaccard (with reference $\beta = 0.85$)



(b) Incremental Jaccard



(c) Rank stability by β

Figure 1: PageRank sensitivity analysis across different teleportation parameters.